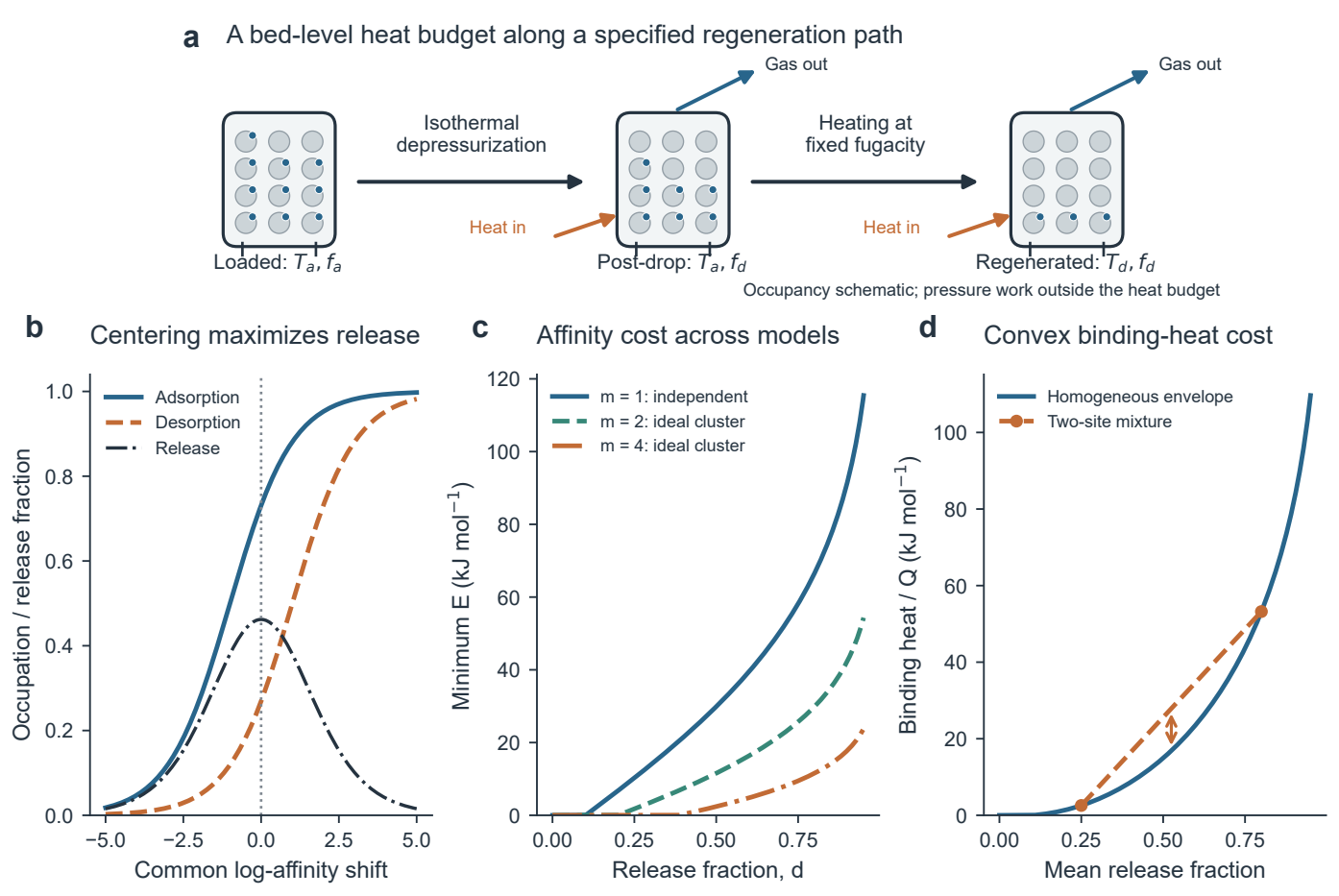

a
A bed-level heat budget along a specified regeneration path
Gas out
Isothermal
depressurization
Heating at
fixed fugacity
Gas out
Heat in
Heat in
Loaded: $T_a$, $f_a$
Post-drop: $T_a$, $f_d$
Regenerated: $T_d$, $f_d$
Occupancy schematic; pressure work outside the heat budget
b
Centering maximizes release
Adsorption
Desorption
Release
Occupation / release fraction
Common log-affinity shift
c
Affinity cost across models
m = 1: independent
m = 2: ideal cluster
m = 4: ideal cluster
Minimum E (kJ mol$^{-1}$)
Release fraction, d
d
Convex binding-heat cost
Homogeneous envelope
Two-site mixture
Binding heat / Q (kJ mol$^{-1}$)
Mean release fraction